9/4/2026

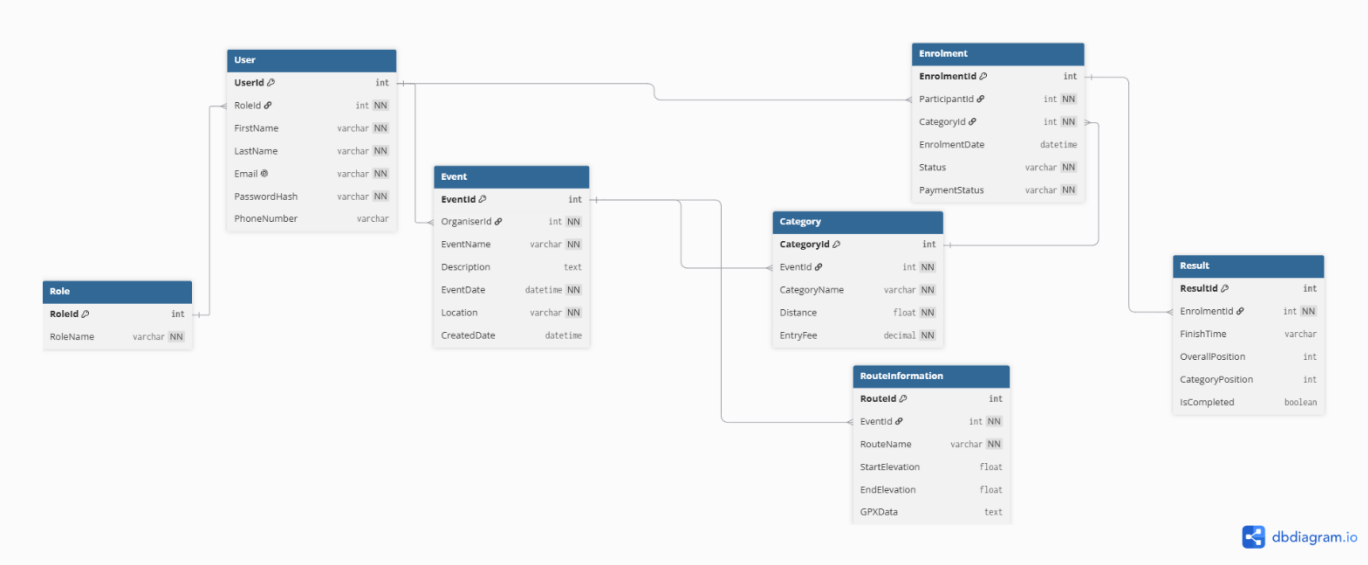
PROGRAMING

ASSIGNMENT 1

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ERD

..



https://dbdiagram.io/d/6a994e75aed2f4f6bea00a93

SQL STATEMENT

```
-- 1. Create Database
CREATE DATABASE RaceDayDB;
GO

USE RaceDayDB;
GO

-- 2. Create Role Table
CREATE TABLE [Role] (
    RoleId INT IDENTITY(1,1) PRIMARY KEY,
    RoleName VARCHAR(50) NOT NULL UNIQUE
);

-- 3. Create User Table
CREATE TABLE [User] (
    UserId INT IDENTITY(1,1) PRIMARY KEY,
    RoleId INT NOT NULL,
    FirstName VARCHAR(100) NOT NULL,
    LastName VARCHAR(100) NOT NULL,
    Email VARCHAR(150) NOT NULL UNIQUE,
    PasswordHash VARCHAR(255) NOT NULL,
    PhoneNumber VARCHAR(20) NULL,
    FOREIGN KEY (RoleId) REFERENCES [Role](RoleId)
);
```

```
-- 4. Create Event Table
CREATE TABLE [Event] (
    EventId INT IDENTITY(1,1) PRIMARY KEY,
    OrganiserId INT NOT NULL,
    EventName VARCHAR(150) NOT NULL,
    Description VARCHAR(MAX) NULL,
    EventDate DATETIME NOT NULL,
    Location VARCHAR(200) NOT NULL,
    CreatedDate DATETIME DEFAULT GETDATE(),
    FOREIGN KEY (OrganiserId) REFERENCES [User](UserId)
);

-- 5. Create Category Table
CREATE TABLE Category (
    CategoryId INT IDENTITY(1,1) PRIMARY KEY,
    EventId INT NOT NULL,
    CategoryName VARCHAR(100) NOT NULL,
    Distance FLOAT NOT NULL,
    EntryFee DECIMAL(10, 2) NOT NULL,
    FOREIGN KEY (EventId) REFERENCES [Event](EventId) ON DELETE CASCADE
);

-- 6. Create RouteInformation Table
CREATE TABLE RouteInformation (
    RouteId INT IDENTITY(1,1) PRIMARY KEY,
    EventId INT NOT NULL,
    RouteName VARCHAR(100) NOT NULL,
```

```
    StartElevation FLOAT NULL,
    EndElevation FLOAT NULL,
    GPXData VARCHAR(MAX) NULL,
    FOREIGN KEY (EventId) REFERENCES [Event](EventId) ON DELETE CASCADE
);

-- 7. Create Enrolment Table
CREATE TABLE Enrolment (
    EnrolmentId INT IDENTITY(1,1) PRIMARY KEY,
    ParticipantId INT NOT NULL,
    CategoryId INT NOT NULL,
    EnrolmentDate DATETIME DEFAULT GETDATE(),
    Status VARCHAR(50) NOT NULL DEFAULT 'Pending',
    PaymentStatus VARCHAR(50) NOT NULL DEFAULT 'Unpaid',
    FOREIGN KEY (ParticipantId) REFERENCES [User](UserId),
    FOREIGN KEY (CategoryId) REFERENCES Category(CategoryId),
    CONSTRAINT UQ_Participant_Category UNIQUE (ParticipantId, CategoryId) --
Prevents duplicate enrolments
);

-- 8. Create Result Table
CREATE TABLE Result (
    ResultId INT IDENTITY(1,1) PRIMARY KEY,
    EnrolmentId INT NOT NULL UNIQUE, -- One-to-One with Enrolment
    FinishTime VARCHAR(20) NULL,
    OverallPosition INT NULL,
    CategoryPosition INT NULL,
```

IsCompleted BIT DEFAULT 1,

FOREIGN KEY (EnrolmentId) REFERENCES Enrolment(EnrolmentId) ON DELETE CASCADE

);

INSERT STATEMENT

USE RaceDayDB;

INSERT INTO [User] (RoleId, FirstName, LastName, Email, PasswordHash, PhoneNumber) VALUES

(1, 'Sipho', 'Dlamini', 'sipho.dlamini@events.co.za', '$2a$12$e8Jk21K91x31/ExampleHashOrg1', '0821234567'),

(1, 'Anika', 'Van Der Merwe', 'anika@sportsmanagement.co.za', '$2a$12$e8Jk21K91x31/ExampleHashOrg2', '0839876543'),

(2, 'Thabo', 'Mokoena', 'thabo.mokoena@gmail.com', '$2a$12$e8Jk21K91x31/ExampleHashPart1', '0711112222'),

(2, 'Sarah', 'Jenkins', 'sarah.j@yahoo.com', '$2a$12$e8Jk21K91x31/ExampleHashPart2', '0723334444');

3. Seed Events (3 Events)

OrganiserId 1 (Sipho), OrganiserId 2 (Anika)

INSERT INTO [Event] (OrganiserId, EventName, Description, EventDate, Location) VALUES

(1, 'Soweto Marathon 2026', 'The iconic Peoples Race through historic Soweto.', '2026-11-01 06:00:00', 'FNB Stadium, Johannesburg'),

(1, 'Cape Town Cycle Tour', 'World famous scenic cycling race around the Cape Peninsula.', '2026-10-15 06:30:00', 'Grand Parade, Cape Town'),

(2, 'Two Oceans Marathon', 'The world standard beautiful ultramarathon and half marathon.', '2026-09-20 05:30:00', 'Newlands, Cape Town');

```
-- 4. Seed Categories (Multiple Categories for Each Event)
-- Event 1: Soweto Marathon (Categories 1, 2, 3)
-- Event 2: Cape Town Cycle Tour (Categories 4, 5)
-- Event 3: Two Oceans Marathon (Categories 6, 7)
INSERT INTO Category (EventId, CategoryName, Distance, EntryFee) VALUES
(1, '10km Open Run', 10.0, 150.00),
(1, '21km Half Marathon', 21.1, 280.00),
(1, '42km Full Marathon', 42.2, 450.00),
(2, '40km Short Route Cycle', 40.0, 350.00),
(2, '109km Full Route Cycle', 109.0, 680.00),
(3, '21km Half Marathon', 21.1, 320.00),
(3, '56km Ultra Marathon', 56.0, 580.00);

-- 5. Seed Route Information
INSERT INTO RouteInformation (EventId, RouteName, StartElevation, EndElevation,
GPXData) VALUES
(1, 'Soweto Classic Route', 1650.0, 1650.0, 'https://raceday.co.za/gpx/soweto2026.gpx'),
(2, 'Peninsula Coastal Loop', 15.0, 15.0,
'https://raceday.co.za/gpx/capetowncycle2026.gpx'),
(3, 'Chapmans Peak Climb', 20.0, 20.0, 'https://raceday.co.za/gpx/twooceans2026.gpx');

-- 6. Seed Enrolments (Sample Enrolments for Participants)
-- ParticipantId 3 (Thabo), ParticipantId 4 (Sarah)
INSERT INTO Enrolment (ParticipantId, CategoryId, Status, PaymentStatus) VALUES
(3, 2, 'Confirmed', 'Paid'),   -- Thabo enrolled in Soweto 21km
(3, 5, 'Confirmed', 'Paid'),   -- Thabo enrolled in Cycle Tour 109km
(4, 1, 'Confirmed', 'Paid'),   -- Sarah enrolled in Soweto 10km
(4, 6, 'Pending', 'Unpaid');   -- Sarah enrolled in Two Oceans 21km
```

-- 7. Seed Results (For Confirmed & Completed Enrolments)

INSERT INTO Result (EnrolmentId, FinishTime, OverallPosition, CategoryPosition, IsCompleted) VALUES

(1, '01:45:30', 142, 28, 1), -- Thabo's result for Soweto 21km

(3, '00:52:10', 85, 12, 1);   -- Sarah's result for Soweto 10km

Table

HTTP Method	**Route**	**Description**	**Role Required**	**Request Body**	**Expected Response**
POST	/api/auth/register	Registers a new user account with a specified role (Organiser or Participant).	None (Public)	{ firstName, lastName, email, password, phoneNumber, roleId }	**201 Created** - User object **400 Bad Request** - Validation error / Email taken
POST	/api/auth/login	Authenticates a user and returns a JWT token containing their user ID and role.	None (Public)	{ email, password }	**200 OK** - Token string & user details **401 Unauthorized** - Invalid credentials

GET	/api/users/profile	Fetches details of the currently logged-in user profile.	Any	None	**200 OK** - Profile data **401 Unauthorized** - Invalid token
PUT	/api/users/profile	Updates profile information for the currently logged-in user.	Any	{ firstName, lastName, phoneNumber }	**200 OK** - Updated user profile **400 Bad Request** - Invalid input
GET	/api/events	Retrieves all upcoming events available for browsing.	None (Public)	None	**200 OK** - Array of event summary objects
GET	/api/events/{id}	Retrieves detailed information for a specific event, including categories and routes.	None (Public)	None	**200 OK** - Detailed event object **404 Not Found** - Event ID does not exist
POST	/api/events	Creates a new road event in the system.	Organiser	{ eventName, description, eventDate, location }	**201 Created** - Created event object **403 Forbidden** - User is

					not an Organiser
PUT	/api/events/{id}	Updates details of an existing event managed by the organiser.	Organiser	{ eventName, description, eventDate, location }	**200 OK** - Updated event object **403 Forbidden** - Not event owner **404 Not Found** - Event does not exist
DELETE	/api/events/{id}	Removes an event from the system along with its categories.	Organiser	None	**204 No Content** - Event deleted **403 Forbidden** - Not event owner **404 Not Found** - Event does not exist
GET	/api/events/{eventId}/categories	Retrieves all distance categories associated with an event.	None (Public)	None	**200 OK** - List of categories **404 Not Found** - Event does not exist

POST	/api/events/{eventId}/categories	Adds a new race distance category to an existing event.	Organiser	{ categoryName, distance, entryFee }	**201 Created** - Category created **403 Forbidden** - User is not an Organiser
POST	/api/enrolments	Registers a participant into a specific event category.	Participant	{ categoryId }	**201 Created** - Enrolment receipt **400 Bad Request** - Already enrolled **403 Forbidden** - User is not a Participant
GET	/api/enrolments/my-enrolments	Retrieves all event enrolments for the logged-in participant.	Participant	None	**200 OK** - Array of participant's enrolments
GET	/api/events/{eventId}/enrolments	Allows an organiser to view all participants enrolled across their event's categories.	Organiser	None	**200 OK** - List of enrolled participants **403 Forbidden** - User is

					not an Organiser
POST	/api/results	Captures and records a participant's finish time and rank for an event.	Organiser	{ enrolmentId, finishTime, overallPosition, categoryPosition, isCompleted }	**201 Created** - Saved result **400 Bad Request** - Result already recorded
GET	/api/results/my-results	Fetches performance history and timing results for the logged-in participant.	Participant	None	**200 OK** - Array of participant results

REFRENCE LIST

Reference List

- **dbdiagram.io.** (2026) *dbdiagram.io: Draw Database Relationship Diagrams*. Available at: https://dbdiagram.io/ [Accessed 3 September 2026].

- **Elmasri, R. and Navathe, S.B.** (2016) *Fundamentals of Database Systems*. 7th edn. Boston: Pearson.
- **Freeman, A.** (2022) *Pro ASP.NET Core 6: Develop Cloud-Ready Web Applications Using MVC, Blazor, and API*. 9th edn. New York: Apress.
- **Microsoft.** (2026) *SQL Server Technical Documentation*. Available at: https://learn.microsoft.com/en-us/sql/sql-server/ [Accessed 3 September 2026].
- **The Independent Institute of Education (IIE).** (2026) *Systems Analysis and Design [Module Manual]*. Johannesburg: The Independent Institute of Education.

Github link

https://github.com/Emmanuel200410/Race-Day-ST10450437